

Build a Churn App with Gradio & Gemini

A step-by-step guide covering slides 14-19: use the Gemini agent in Google Colab to run an end-to-end data science project on the Telco churn data - from data understanding to model building, evaluation, and a deployed Gradio app.

Activity is designed by Su Nguyen; Detailed guide is drafted by Claude Cowork.

Before you start

You need a Google account (for Colab). Use the Telco-Customer-Churn.csv file from last week. Everything runs in your browser - no installation required. If you need a refresher on opening Colab and using the Gemini agent, see the Week 2 and Week 3 walkthroughs.

Slide 15	Upload the CSV and give the agent one end-to-end prompt
Slide 16	Read the AI's auto-generated plan
Slide 17	Pause and review the plan before running
Slide 18	Test the deployed Gradio app
Slide 19	Improve the app: sliders + batch evaluation with metrics

Tip: dock the Gemini agent panel to the right side of Colab (drag it or use the panel's expand control) so you can see the notebook and the agent's plan side-by-side.

Step 1 - Upload the data and prompt the agent (Slide 15)

Open a new Colab notebook and upload Telco-Customer-Churn.csv via the Files panel (folder icon). Open the Gemini agent and dock it to the right. Then give it one comprehensive prompt:

```
You are a professional data scientist. Help me load the
Telco-Customer-Churn.csv file, examine what columns are included.
The goal is to develop machine learning models to predict Churn.
Perform the following high-level steps: (1) data understanding
(2) model building (build at least 2 interpretable ML models such as
logistic regression or decision trees and one complex one like
gradient boosting) (3) deploy the models for internal testing using Gradio.
```

Tip: dock the agent panel to the right so the notebook and the plan are visible together.

Step 2 - Read the AI's plan (Slide 16)

Instead of writing code immediately, the agent first lays out a plan. Yours may differ slightly, but it should cover the full pipeline:

1. Load and Examine Data - read the CSV, show rows, types, missing values.
2. Preprocess Data for Modeling - encode categoricals, scale, handle missing values.
3. Split Data - train/test split for robust evaluation.
4. Train Interpretable Models - e.g. Logistic Regression and Decision Tree.
5. Train Complex Model - e.g. Gradient Boosting Classifier.
6. Evaluate Model Performance - accuracy, precision, recall, F1, ROC-AUC, and compare models.
7. Deploy Models with Gradio - an interactive interface for internal testing.
8. Final Task - summarise findings and insights.

 You are a professional data scientist. Help me load the Telco-Customer-Churn.csv file, examine what columns are included. The goal is to develop machine learning models to predict Churn. Perform the following high-level steps: (1) data understanding (2) model building (build at least 2 interpretable ML models such as logistics regression or decision trees and one complex one like gradient boosting) (2) deploy the models for internal testing using Gradio.

◆ Great! I've outlined a comprehensive plan to tackle your Telco customer churn prediction project, from data understanding to model deployment. Take a look and let me know if any adjustments are needed.

- **Load and Examine Data:** Load the 'Telco-Customer-Churn.csv' file into a pandas DataFrame. Display the first few rows, column names, data types, and check for missing values to understand the dataset's structure and quality.
- **Preprocess Data for Modeling:** Perform necessary data preprocessing steps, including handling categorical variables (e.g., one-hot encoding), scaling numerical features if required, and addressing any missing values. This will prepare the data for training machine learning models.
- **Split Data:** Divide the preprocessed dataset into training and testing sets to ensure robust model evaluation.
- **Train Interpretable Models:** Train at least two interpretable machine learning models, such as Logistic Regression and Decision Tree Classifier, on the training data to predict 'Churn'.
- **Train Complex Model:** Train a complex machine learning model, such as Gradient Boosting Classifier, on the training data to predict 'Churn'.
- **Evaluate Model Performance:** Evaluate the performance of all trained models using appropriate metrics (e.g., accuracy, precision, recall, F1-score, ROC-AUC) on the test set. Provide a comparison of the model performances.
- **Deploy Models with Gradio:** Develop a Gradio interface to deploy the trained models, allowing for interactive testing and prediction. This interface will be used for internal testing purposes.
- **Final Task:** Summarize the findings from the data analysis, the performance of the developed models, and the insights gained from the project.

Generating...

▸ Preparing

🔊 Load and Examine Data

🔊 Preprocess Data for Modeling

🔊 Split Data

🔊 Train Interpretable Models

🔊 Train Complex Model

🔊 Evaluate Model Performance

🔊 Deploy Models with Gradio

🔊 Final Task

Figure 1 - The agent's auto-generated plan for the end-to-end churn project.

Step 3 - Pause and review the plan (Slide 17)

Before running, read the plan critically - this is the point of the activity. Ask yourself:

- Are you happy with the plan?
- Is it clear enough, or would any step benefit from more detailed instructions?
- Do you think you'll get the outputs you want?

Revise the prompt and re-run if needed. When you're satisfied, let the agent run (auto-run / Accept & Run). It will work through each step, and may pause for you to Accept & Run each stage.

Step 4 - Test the Gradio app (Slide 18)

After the models are trained and evaluated, the agent builds and launches a Gradio app inside Colab. Enter a customer's details and click Submit to get a churn prediction - here a customer is predicted NOT to churn with a probability of 0.55.

Telco Customer Churn Predictor

Enter customer details to predict churn likelihood.

Gender Female	B. output Customer is predicted NOT to churn with a probability of 0.55 Flag
Senior Citizen 0	
Partner No	
Dependents No	
Tenure (months) 1	
Phone Service No	
Multiple Lines No phone service	
Internet Service DSL	
Online Security No	
Online Backup	

Figure 2 - The deployed Gradio app: enter customer details, get a churn prediction.

Discuss: what is the app used for? Is it helpful for internal testing? Which ML model powers it? How could you improve it?

Step 5 - Improve the app (Slide 19)

Refine the app with a follow-up prompt so it is easier to use and supports proper evaluation:

```
Improve the gradio app by (1) using sliders for numeric variables  
(2) allow user to upload testing dataset and visualise the  
performance metrics with testing data
```

Now the app has a Batch Evaluation tab: upload a testing CSV and it reports the model's metrics - here Accuracy 0.83, Precision 0.68, Recall 0.54, F1 0.60 and ROC-AUC 0.89.

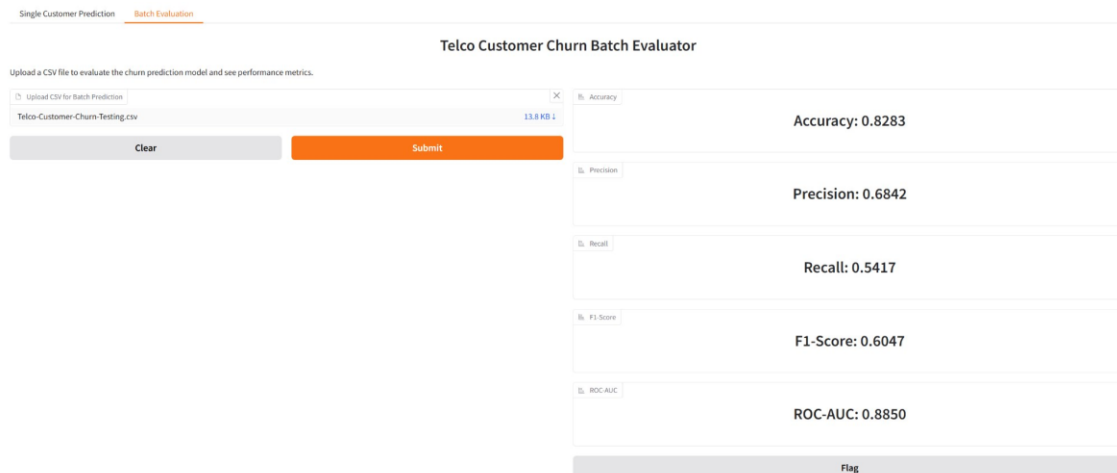


Figure 3 - The improved app: upload a test set and see Accuracy, Precision, Recall, F1 and ROC-AUC.

Good to know

- Always review AI-generated code before trusting it. Colab warns: Gemini can make mistakes so double-check it and use code with caution.
- The agent works in stages - it may pause and ask you to Accept & Run each batch. Let each stage finish before continuing.
- Auto-run mode runs the generated cells automatically; you can turn it off if you prefer to inspect each cell first.
- Uploaded files and the runtime are temporary; if the session drops, re-upload the CSV and re-run the cells.
- For Assessment 1: document the exact prompts you used (with your assumptions and expected outcomes) and explain how you verified the AI's outputs.